

HDD-ISA: AI Architectures for Causal Discriminations

An Interface Specification for Testing Memory, Recurrence, Self-Reference, and Self-Modeling

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Companion specification to Taotuner (2026), *History-Dependent Dynamics (HDD): A Methodological Framework for Disentangling History Dependence, Recurrence, Self-Reference, and Self-Modeling in Dynamical Systems*.

Abstract

Modern AI architectures increasingly incorporate persistent state, recurrence, predictive representations, agent representations, and mechanisms capable of representing aspects of the system itself. The presence of such mechanisms, however, does not establish the functional role attributed to them. Likewise, behavioral effects alone generally do not identify the internal mechanism responsible for those effects.

We propose HDD-ISA, an architectural interface specification for translating the five functional constructs defined by History-Dependent Dynamics (HDD) into experimentally addressable state variables, intervention ports, outcome measures, and control conditions.

HDD-ISA has two complementary applications. In retrofit mode, it provides a framework for instrumenting existing architectures so that HDD constructs can be tested post hoc. In design mode, it specifies architectural interfaces that permit causal discrimination from the outset.

The framework distinguishes five inferential stages: architectural access, intervention validity, target engagement, causal effect, and hypothesis discrimination. It further defines construct-specific intervention interfaces, a separation-matrix taxonomy for cross-effects and identification-compromising leakage, implementation protocols for recurrent and transformer-based architectures, and explicit revision and falsification criteria.

The central methodological claim is:

HDD-ISA does not determine whether an architecture possesses a functional construct. It specifies the architectural interfaces and experimental conditions under which competing hypotheses about that construct become causally distinguishable.

Keywords: history dependence; recurrence; self-reference; self-modeling; causal identification; intervention separability; target engagement; AI architecture; falsifiability.

1. Introduction

Modern AI architectures increasingly contain persistent state, recurrent computation, predictive models, agent representations, and mechanisms capable of representing aspects of the system itself. These capabilities make it possible for an architecture to implement functions that are naturally described using terms such as memory, recurrence, self-reference, and self-modeling.

Architectural presence alone, however, does not establish the corresponding functional construct.

A system may contain a persistent state without using historical information predictively. It may exhibit history-dependent behavior without the relevant effect being mediated by the particular memory variable

under investigation. A recurrent architecture may generate recurrence-like input-output behavior through mechanisms other than the feedback pathway of interest. A model may manipulate representations labeled "self" without functionally referring to itself. Likewise, a system may accurately predict or describe its own future without using that representation to determine its actions.

These possibilities create a methodological identification problem.

The relevant question is therefore not simply:

Does the architecture contain a mechanism resembling the construct?

but rather:

Can competing functional explanations of that mechanism be causally discriminated?

The History-Dependent Dynamics (HDD) framework (Taotuner, 2026) defines five diagnostic constructs:

1. History-dependent prediction;
2. Causal trajectory dependence;
3. Feedback recurrence;
4. Functional self-reference;
5. Self-modeling.

These constructs are independent functional claims. HDD does not treat them as a developmental hierarchy, nor does it make ontological claims about consciousness or subjective experience.

HDD specifies the evidential conditions under which these constructs may be supported. It does not, by itself, specify what an AI architecture must expose for those conditions to be experimentally testable.

HDD-ISA addresses this architectural layer. We refer to the inferential distance between implementing a candidate mechanism and establishing its functional role as the HDD Gap — a gap between:

having a mechanism

and:

knowing, with causal evidence, what functional role that mechanism plays.

Its central question is:

What must an AI architecture expose, or be designed to expose, so that HDD constructs can be causally tested rather than merely inferred from behavioral performance?

The answer is expressed in terms of state variables, intervention ports, target-engagement measurements, outcome variables, and controls.

HDD-ISA therefore supports two modes of use:

- Retrofit: instrumenting an existing architecture to determine which HDD constructs are experimentally accessible.

- Design: constructing architectures whose relevant mechanisms are independently addressable and whose cross-effects can be characterized.

The framework does not certify what an architecture is. It specifies the experimental interfaces required for competing hypotheses about its functional organization to become distinguishable.

The fundamental distinction is:

$$\text{mechanism} \not\Rightarrow \text{functional interpretation}$$

and, conversely:

$$\text{functional effect} \not\Rightarrow \text{unique mechanism}$$

Causal interpretation therefore requires a chain:

$$\text{architectural access} \rightarrow \text{intervention validity} \rightarrow \text{target engagement} \rightarrow \text{causal effect} \rightarrow \text{hypothesis discrimination}$$

Failure at any stage constrains the conclusions available at subsequent stages.

2. HDD Functional Constructs

HDD defines five constructs. Each represents a distinct functional claim and therefore requires distinct evidence.

Construct	Question	Core Evidence
C-I	Does historical information improve prediction?	Predictive gain from historical information beyond the present state
C-II	Does the prior trajectory causally affect the future?	Trajectory intervention changes future outcomes under matched present state
C-III	Is that effect mediated by a specific feedback pathway?	Selective disruption of the candidate feedback pathway changes the trajectory effect
C-IV	Does the system causally use self-relevant information?	Differential response to referent substitution under matched controls
C-V	Does the system model its own future and use that model?	Self-prediction, counterfactual self-prediction, and causal use in policy selection

HDD-ISA does not redefine these constructs. It specifies the architectural conditions required to test them.

3. HDD-ISA Architectural Requirements

3.1 Intervention Separability

Intervention separability is not treated as an intrinsic scalar property of an architecture. It is a property of the architecture together with the hypotheses, interventions, measurements, and controls available to the experiment.

Define the experimental tuple:

$$(A, H, I, Y, C)$$

where:

- *A is the architecture;*
- *H is the competing hypothesis set;*
- *I is the available intervention set;*
- *Y is the measurable outcome set;*
- *C is the available control set.*

The same architecture can therefore be identifiable under one experimental interface and non-identifiable under another.

3.2 Interface Testability

Let C denote an HDD construct. Define interface testability as:

$$T_{interface_C}(A; I, Y, C) = 1$$

iff there exists an intervention I_C , outcome set Y_C , and control condition C_C that are jointly implementable in architecture A.

This criterion establishes only that the experiment can be performed. It does not establish that the intended mechanism has been engaged or that competing hypotheses can be distinguished.

3.3 Hypothesis Discrimination

For a hypothesis H, define its intervention signature as the family of conditional distributions indexed by interventions and outcomes:

$$S_{\{I,Y\}}(H; A) = \{ P_H(Y | do(I), A) \} \text{ for } (I,Y) \in I \times Y$$

Two hypotheses are interventionally equivalent when:

$$H_a \sim_{\{I,Y,A\}} H_b \text{ iff } S_{\{I,Y\}}(H_a; A) = S_{\{I,Y\}}(H_b; A)$$

Define hypothesis discrimination as:

$$Disc(H_a, H_b | A, I, Y) = 1 \text{ iff } S(H_a; A) \neq S(H_b; A)$$

Construct-level discrimination therefore requires more than an available intervention port.

We require:

$$T_{disc_C} = 1$$

iff:

6. the intervention, outcomes, and controls are implementable;
7. the intervention can be validated;
8. target engagement can be independently assessed;

9. at least two relevant hypotheses in H_C predict distinguishable intervention signatures.

Thus:

$$\text{interface existence} \not\Rightarrow \text{hypothesis discrimination}$$

An architecture can consequently possess a nominal intervention interface while remaining non-identifiable under the current experimental design.

4. The Five-Stage Causal Chain

HDD-ISA separates five stages.

Stage	Meaning
1. Architectural access	The relevant state or pathway can be accessed or manipulated
2. Intervention validity	The manipulation is an empirically adequate implementation of the causal operator specified by the hypothesis, within prespecified invariants
3. Target engagement	Independent evidence confirms that the hypothesized target was altered
4. Causal effect	The intervention changes the specified outcome under valid controls
5. Hypothesis discrimination	The resulting signature distinguishes the target hypothesis from relevant alternatives

The complete inference chain is:

$$\text{access} \rightarrow \text{validity} \rightarrow \text{engagement} \rightarrow \text{effect} \rightarrow \text{discrimination}$$

A positive result at one stage does not automatically establish the next.

5. Construct-to-Interface Mapping

Construct	Required Architectural Interface	Primary Intervention	Does Not Establish
C-I	Persistent historical state M	$\text{do}(M = \emptyset)$ or placebo-history intervention	Trajectory causality, recurrence, self-reference
C-II	Trajectory–state decoupling	$\text{do_gen}(T = \tau^A)$ vs. $\text{do_gen}(T = \tau^B)$ under matched present state	That memory M mediates the effect
C-III	Independently bypassable feedback pathway R	$\text{do}(R = 0)$ vs. capacity-matched alternative	Recurrence in every computational sense
C-IV	Manipulable agent-referent variable/interface	$\text{do}(S_{\text{self}})$ vs. $\text{do}(S_{\text{ext}})$	Selfhood, agency, consciousness
C-V	Self-state S, self-model SM, causal pathway $SM \rightarrow PA$	Self-Model Causal Intervention	Self-modeling from prediction or description alone

5.1 C-I: Memory Intervention

The intervention $\text{do}(M = \emptyset)$ is not automatically interpretable as "the causal value of memory." Δ_I is interpretable only after the placebo-history intervention has demonstrated equivalence in non-historical dimensions (capacity, activation distribution, information throughput, latency). Without such validation, an observed effect may simply reflect generic capacity loss.

5.2 C-II: Two Distinct Trajectory Operators

For C-II, two distinct causal operators must be distinguished:

- $\text{do_replay}(\tau)$: intervention on the observed trajectory realization (replaying a sequence)
- $\text{do_gen}(T = \tau)$: intervention on the causal components that generate the trajectory

Only the latter constitutes a proper causal intervention for C-II. Replaying an observed sequence should not be treated as equivalent unless the relevant generative states and transition dynamics are shown to be preserved.

For C-II, Z_n denotes the present-state variables that competing hypotheses require to be matched. Let:

$$d_Z(Z_n^A, Z_n^B) \leq \varepsilon$$

define admissible matching, where both d_Z and ε are prespecified.

Approximate matching changes the estimand: the resulting quantity is a conditional trajectory effect, not necessarily the total causal effect of history.

Because Z_n may lie downstream of the manipulated trajectory, conditioning on it can block part of the causal pathway:

$$\text{null conditional trajectory effect} \not\Rightarrow \text{absence of total historical effect}$$

5.3 C-III: Structural vs. Functional Recurrence

C-III distinguishes two concepts:

$$R_{\text{structural}} \neq R_{\text{functional}}$$

- $R_{\text{structural}}$: the presence of a computational feedback pathway
- $R_{\text{functional}}$: the causal role that pathway plays in the history-dependent effect

A Transformer may exhibit temporal dependencies without implementing the feedback dynamics relevant to C-III. Conversely, a system may implement a functionally equivalent effect without an explicit loop.

C-III establishes the latter, not merely the former.

6. Architectural Compatibility Levels

HDD-ISA distinguishes three levels.

Level	Criterion
HDD-Compatible	At least one construct has a testable interface

Level	Criterion
HDD-Interface-Complete	All five constructs have testable interfaces
HDD-Designed	All five constructs have independently addressable interfaces and experimentally characterized cross-effects

$$HDD-Compatible \Leftrightarrow \exists C : T interface_C = 1$$

$$HDD-Interface-Complete \Leftrightarrow \forall C : T interface_C = 1$$

HDD-Designed additionally requires the design rules defined below and characterization of the separation matrix.

These labels describe experimental capability, not evidential success. An HDD-Interface-Complete architecture with severe intervention leakage may provide weaker causal evidence for a specific construct than an HDD-Compatible architecture with a rigorously validated intervention for that one construct.

7. Design Rules

DR1. Every state variable or pathway whose causal role is required by a tested HDD hypothesis must have a dedicated intervention port or an experimentally validated equivalent intervention.

DR2. Every intervention port must support matched controls preserving relevant surface, capacity, task, and informational properties. Placebo interventions must be prespecified independently of observed outcomes.

DR3. A self-model representation must possess an experimentally addressable causal pathway into action or policy selection.

DR4. A feedback pathway under investigation must be independently bypassable without disabling general system function. Capacity-preservation must be empirically tested against a set of prespecified operational invariants.

DR5. Historical and present-state variables must be experimentally distinguishable under the available intervention set. Induced changes in non-target variables must be measured rather than assumed absent.

DR6. Self and external referents must be representable through a common manipulable interface, allowing referent identity to change while task structure is preserved.

DR7. Cross-effects among intervention ports must be experimentally characterized and reported.

DR8. Pathway ablation does not by itself establish pathway-specific causal evidence without a matched control for equivalent computational degradation.

DR9. Functional separability does not require architectural modularity; causal addressability is sufficient.

DR10. A claim of structural recurrence ($R_{\text{structural}}$) does not establish functional feedback recurrence ($R_{\text{functional}}$); the latter requires causal evidence specific to the history-dependent effect.

DR11. For C-I, $\text{do}(M = \emptyset)$ is not interpretable as causal value of memory without validating that the observed effect is not simply generic capacity loss.

These rules specify conditions for causal identification. They do not require literal physical modularity.

8. Separation Matrix and Intervention Leakage

An intervention targeting one construct may alter variables relevant to another. Such cross-effects must be distinguished from identification-compromising leakage.

For intervention I_i and outcome Z_j , define:

$$X_{ij} = D(P(Z^j) / do(I_i), P(Z^j) / do(I_i^0)))$$

where D is a prespecified distance or divergence, Z^j is a target variable relevant to construct j , and I_i^0 is a prespecified placebo intervention that reproduces the non-target perturbation of I_i .

Placebo prespecification: I_i^0 must be defined before outcome analysis and validated on the same non-target perturbation dimensions used to define the corresponding intervention.

A nonzero X_{ij} establishes a cross-effect. It does not by itself establish leakage.

Define identification-compromising leakage:

$$L_{ij} = 1$$

when the cross-effect caused by I_i materially compromises identification of the causal contribution attributed to construct i .

Cross-effect vs. confounding: A cross-effect may be perfectly known and modelable. It becomes a problem only when it prevents identification of the relevant effect. Known coupling is not non-identification.

Therefore:

$$\text{cross-effect} \not\Rightarrow \text{leakage}$$

and:

$$\text{known coupling} \not\Rightarrow \text{non-identification}$$

Each matrix cell should be classified as:

Classification	Meaning
Target	Expected effect on the construct under investigation
Permitted coupling	Expected cross-effect that does not compromise interpretation
Leakage	Cross-effect that compromises identification
Uncharacterized	Not yet empirically measured
Non-identifiable	Available interface cannot distinguish competing effects

A canonical reporting matrix is:

Intervention	C-I	C-II	C-III	C-IV	C-V
M	Target	Uncharacterized	Uncharacterized	Uncharacterized	Uncharacterized
τ	Permitted	Target	Uncharacterized	—	Uncharacterized
R	—	—	Target	—	Uncharacterized
S	—	—	—	Target	Permitted
SM	—	—	—	Permitted	Target
PA	—	—	—	—	Target

Here, "—" denotes a theoretically excluded pathway under the architecture's stated causal model. It is not empirical evidence of zero effect.

All remaining non-target cells are Uncharacterized until measured.

The separation matrix is therefore more informative for causal claims than a modular architecture diagram alone:

$$\text{module separability} \not\Rightarrow \text{causal separability}$$

9. Target Engagement Protocol

An intervention port is not itself evidence that the intended target has been manipulated.

$$\text{port manipulation} \not\Rightarrow \text{target engagement}$$

Intervention validity and target engagement are distinct stages:

- Intervention validity establishes that the manipulation is an empirically adequate implementation of the causal operator specified by the hypothesis.
- Target engagement establishes that the hypothesized target was altered.

Nominal vs. real interventions: Distinguish I_{nominal} (the intervention the experimenter intends to perform) from I_{real} (the transformation actually produced). Validity requires evidence that $I_{\text{real}} \approx I_{\text{causal}}$ within a prespecified set of invariants.

HDD-ISA requires independent target-engagement evidence. A claim of target engagement is supported when the following four criteria are satisfied:

TE1 — Manipulation Fidelity: The intervention produces the predicted change in the target variable or pathway.

TE2 — Target Specificity: The change is distinguishable from non-specific perturbations.

TE3 — Invariant Preservation: Relevant architectural invariants (capacity, task performance, information throughput) remain within prespecified bounds.

TE4 — Replication: The engagement pattern is reproducible across independent trials.

The inferential rule is:

unconfirmed target engagement $\Rightarrow$ no construct inference

A failed intervention should therefore be classified as UE, not automatically as negative evidence.

10. Construct Identification vs. Mechanism Identification

HDD-ISA distinguishes two forms of identification that are often conflated.

Construct identification establishes that a functional construct is supported by the intervention signature. For example, C-IV may be supported by evidence that behavior is causally sensitive to referent identity.

Mechanism identification establishes which specific internal implementation produces that effect. Multiple mechanisms may be compatible with the same construct.

Thus:

construct identification $\nRightarrow$ mechanism identification

and:

mechanism implementation $\nRightarrow$ construct identification

This yields a hierarchy of inferential specificity:

behavioral sensitivity $<$ construct identification $<$ mechanism identification

This is not an ontological hierarchy. It is a hierarchy of evidential specificity.

11. Construct-Specific Protocols

C-I — History-Dependent Prediction

Requirement: persistent historical state M.

Intervention: $\text{do}(M = \emptyset)$ or a placebo-history manipulation $\text{do}(M = M_{\text{placebo}})$ where the placebo preserves relevant dimensionality and generic capacity while disrupting temporal information.

Estimand:

$$\Delta_I = L_{\text{no-memory}} - L_{\text{memory}}$$

Evidence: $\Delta_I > 0$ under prespecified capacity and task controls.

Interpretation: A positive result establishes predictive utility of historical information. It does not establish trajectory causality, recurrence, self-reference, or self-modeling. $\text{do}(M = \emptyset)$ is not interpretable as causal value of memory without validating that the observed effect is not simply generic capacity loss.

C-II — Causal Trajectory Dependence

Requirement: trajectory–state decoupling.

Construct trajectories $T^A \neq T^B$ such that $d_Z(Z_n^A, Z_n^B) \leq \epsilon$.

Define $M = \{ Z : d_Z(Z_n^A, Z_n^B) \leq \epsilon \}$.

The conditional trajectory effect is:

$$\Delta_{II}(h) = E[Q(O_{\{n+h\}}) \mid do_gen(T=\tau^A), Z_{n \in M}] - E[Q(O_{\{n+h\}}) \mid do_gen(T=\tau^B), Z_{n \in M}]$$

Operator clarification: $do_gen(T = \tau)$ denotes intervention on the causal components that generate the trajectory. $do_replay(\tau)$ (replaying an observed sequence) should not be treated as equivalent unless the relevant generative states and transition dynamics are shown to be preserved.

Interpretation: A positive result establishes that differentiated trajectories retain causal influence beyond the specified matched present-state variables. It does not establish that a particular memory variable mediates the effect:

$$trajectory\ causality \not\Rightarrow memory\ causality$$

C-III — Feedback Recurrence

Requirement: independently bypassable feedback pathway R and a capacity-matched comparison condition.

Let $R \in \{0,1\}$ indicate feedback absence/presence. Let $K \in \{0,1\}$ indicate a paired control intervention condition:

- $K=1$: computational capacity is preserved (operationalized by a set of invariants: performance, latency, activation statistics, throughput)
- $K=0$: controlled degradation condition

K is not a primitive causal variable; it is an experimental condition that must be empirically validated.

The interaction estimand is:

$$\Delta_{III} = (Y_{\{R=1, K=1\}} - Y_{\{R=1, K=0\}}) - (Y_{\{R=0, K=1\}} - Y_{\{R=0, K=0\}})$$

The relevant hypothesis is:

$$H0: \Delta_{III} \leq 0 \text{ versus } H1: \Delta_{III} > 0$$

Structural vs. functional recurrence: $R_{\text{structural}}$ (presence of a feedback pathway) does not establish $R_{\text{functional}}$ (causal role in history-dependent effect). C-III establishes the latter.

Interpretation: The tested historical effect is specifically modulated by the candidate feedback pathway. It should not be inflated into the broader claim that "the architecture is recurrent."

Control requirement: The capacity-matched condition must be empirically validated against the prespecified invariants. Pathway ablation does not establish pathway-specific causal evidence without such validation.

C-IV — Functional Self-Reference

Requirement: manipulable agent-referent variable/interface. The term "agent-referent" is used rather than "self-referent" because the functional status of the representation is precisely what is being tested.

The referent should be switchable between S_{self} and S_{ext} while preserving task structure and relevant instrumental properties.

A basic 2×2 design manipulates referent and truth.

The interaction is:

$$\Delta_{IV} = [P(A/S_T^{self}) - P(A/S_F^{self})] - [P(A/S_T^{ext}) - P(A/S_F^{ext})]$$

Evidence: $\Delta_{IV} \neq 0$ together with referent sensitivity, invariance to surface-form changes, instrumental equivalence, and replication across task types.

The inference remains strictly functional:

$$self-reference \not\Rightarrow selfhood \not\Rightarrow consciousness$$

C-V — Self-Modeling

Requirement: S, SM, SM $\rightarrow$ PA where S is system state, SM is the self-model, and PA is the policy-access pathway.

C-V consists of three dimensions, each with an explicit operational criterion:

V1 — Self-prediction: The system predicts relevant aspects of its own future state with out-of-sample accuracy exceeding a prespecified baseline. Verbal description or self-report alone is insufficient; prediction must be evaluated on held-out trajectories.

V2 — Counterfactual self-prediction: The system predicts how its own future state would differ under counterfactual conditions. This requires sensitivity to interventions on the system's own state, evaluated against independently generated counterfactual rollouts.

V3 — Causal model use: Intervention on self-model content changes policy selection through the SM $\rightarrow$ PA pathway. The effect must be present when PA is open and attenuated when PA is blocked.

Epistemic vs. instrumental distinction: V1 and V2 establish properties of model adequacy (self-prediction and counterfactual competence). V3 establishes instrumental causal use. Self-modeling requires both:

$$self-modeling = self-prediction + counterfactual competence + causal policy use$$

Critical requirement: V1 $\wedge$ V2 $\wedge$ V3 must establish that the representation identified as SM is both predictive of the system's future states and causally used in policy selection. Correlation among V1, V2, and V3 is insufficient.

Therefore:

$$V = V1 \wedge V2 \wedge V3$$

V1 alone is insufficient.

Self-Model Causal Intervention (SMCI): SMCI directly manipulates self-model content: $do(\hat{S}_{self} = \hat{s})$. The manipulation is evaluated under two conditions: PA = open and PA = blocked.

The estimand is:

$$\Delta_V = [E(Y|do(\hat{S}=\hat{s}_{true})) - E(Y|do(\hat{S}=\hat{s}_{false}))]_{\{PA=open\}} - [E(Y|do(\hat{S}=\hat{s}_{true})) - E(Y|do(\hat{S}=\hat{s}_{false}))]_{\{PA=blocked\}}$$

Control requirement: PA = blocked must be validated with capacity-preservation controls to ensure that the attenuation is not simply generic degradation.

The critical signature is an intervention-sensitive effect that exists when the policy pathway is available and is attenuated when that pathway is blocked.

Thus:

$$\textit{self-description} \not\Rightarrow \textit{self-model use}$$

and:

$$\textit{self-prediction} \not\Rightarrow \textit{self-modeling}$$

unless causal model use is demonstrated.

12. Implementation Guide

An implementation should proceed construct by construct.

For each construct, first identify whether the required architectural interface exists. If it does not, the construct should be explicitly reported as architecturally untestable under the current interface:

$$T \textit{interface_C} = 0$$

It should not silently disappear from the analysis.

For a transformer-based agent, for example:

Construct	Candidate Component	Intervention
C-I	Context/KV state	$\text{do}(M = \emptyset)$
C-II	Observation/action trajectory	$\text{do_gen}(T=\tau^A)$ vs. $\text{do_gen}(T=\tau^B)$
C-III	Explicit recurrent pathway, if present	$\text{do}(R = 0)$
C-IV	Agent-referent variable/interface	$\text{do}(S_{\text{self}})$ vs. $\text{do}(S_{\text{ext}})$
C-V	Dedicated self-model representation	SMCI

A standard feedforward transformer does not automatically provide a C-III interface. A recurrence claim therefore requires an explicit feedback pathway whose manipulation can be distinguished from general computational degradation.

For an RNN/LSTM:

Construct	Candidate Component	Intervention
C-I	Hidden/history state	$\text{do}(M = \emptyset)$
C-II	Hidden-state trajectory	$\text{do_gen}(T=\tau^A)$ vs. $\text{do_gen}(T=\tau^B)$
C-III	Recurrent transition	$\text{do}(R = 0)$
C-IV	Agent-referent representation	Self/external substitution
C-V	Self-model readout	SMCI with PA-open/blocked

For black-box LLM deployments, prompt manipulations can serve as behavioral probes but should not automatically be described as architectural interventions.

Thus:

$$\text{prompt probe} \not\Rightarrow \text{architectural intervention}$$

unless the prompt manipulation has been independently linked to the internal variable specified by the hypothesis.

13. Controls and Confounders

Construct	Principal Alternative	Required Control
C-I	Capacity rather than history	Capacity-matched comparison and temporal placebo
C-II	Latent present-state differences	Prespecified trajectory–state matching
C-III	General computational damage	Capacity-matched feedforward baseline with validated invariants
C-IV	Label association or pragmatic inference	Referent swapping and relevance controls
C-V	Generic degradation	Capacity-matched manipulation and PA-blocked control

Where feasible, controls should preserve task difficulty, computational capacity, information quantity, surface complexity, reward structure, temporal position, and generic uncertainty.

14. Reporting States

HDD-ISA recommends four principal inferential states.

State	Interpretation
Supported	Target engagement confirmed, controls pass, and the observed signature discriminates the target hypothesis
Negative Evidence	Target engagement and intervention validity confirmed, but the predicted signature is absent
UE — Uninterpretable	Intervention validity or target engagement was not adequately established
NI — Non-Identifiable	The experiment was successfully executed, but available interventions and outcomes leave competing hypotheses equivalent

This distinction prevents two important inferential errors:

$$UE \not\Rightarrow \text{evidence against}$$

and:

$$NI \not\Rightarrow \text{evidence of absence}$$

15. Identifiability Under Intervention Budget

In practice, an intervention may be theoretically sufficient but experimentally infeasible due to cost, system destruction, inaccessibility, or excessive degradation.

Distinguish:

- Non-identifiable: no available intervention set can separate the hypotheses
- Not practically identifiable under budget B: the intervention set exists but is infeasible under practical constraints

Thus:

$$NI \not\Rightarrow \text{practically infeasible}$$

and:

$$\text{practically infeasible} \not\Rightarrow \text{in principle impossible}$$

This limitation should be considered when interpreting results.

16. The HDD Gap as an Identification Problem

The HDD Gap is the inferential distance between implementing a candidate mechanism and establishing its functional role. It consists of two distinct identification problems.

Forward Gap (G_forward):

$$\text{mechanism implemented} \not\Rightarrow \text{functional hypothesis established}$$

This is the problem of functional interpretation: given that a mechanism is present, what functional construct, if any, does it implement?

Inverse Gap (G_inverse):

$$\text{functional effect observed} \not\Rightarrow \text{unique mechanism identified}$$

This is the problem of mechanistic identification: given an observed functional effect, which internal mechanism produced it?

These two gaps correspond to two distinct fallacies:

Implementation fallacy: "A mechanism resembling X exists, therefore the system implements X."

Reverse-mechanism fallacy: "Behavior compatible with X is observed, therefore we know which internal mechanism produced it."

These gaps are logically independent. An architecture may contain a genuine mechanism whose causal role cannot be identified with the available interventions. Conversely, an observed functional effect may be genuine while several distinct mechanisms remain compatible with the data.

The HDD Gap is therefore not a single problem but two:

$$\begin{aligned} \text{mechanism} &\text{ —}G_{\text{forward}}\not\Rightarrow\text{ — construct} \\ \text{effect} &\text{ —}G_{\text{inverse}}\not\Rightarrow\text{ — mechanism} \end{aligned}$$

HDD-ISA narrows both gaps by expanding the experimental interface:

$$(I, Y) \rightarrow (I', Y')$$

with the explicit objective of separating competing explanations.

17. Scope and Limitations

HDD-ISA is explicitly interface-relative.

1. It discriminates only among hypotheses included in H. It cannot rule out mechanisms that were never considered.
2. Interventions may perturb multiple mechanisms. The separation matrix characterizes this problem but cannot guarantee perfect isolation.
3. Representations are generally non-unique. The same input-output behavior may be implemented through multiple internal mechanisms.
4. Demonstrations may be distribution-dependent. Claims involving self-reference and self-modeling require replication across tasks.
5. C-II relies on approximate present-state matching. The resulting estimand is conditional on the matching procedure.
6. Black-box prompt experiments remain behavioral probes unless their relationship to the hypothesized internal mechanism is independently established.
7. A hypothesis pair may be non-identifiable under one interface and identifiable under another:

$$NI(A, H, I1, Y1) \not\Rightarrow NI(A, H, I2, Y2)$$

8. Identifiability may be infeasible under practical intervention budgets even when theoretically possible.

18. Revision and Falsification Criteria

If a construct remains systematically non-identifiable across an architecture class despite reasonable intervention and measurement designs, it should be classified as NI for that architecture class rather than repeatedly tested with inadequate interfaces.

If target engagement cannot be independently established for a class of architectures, claims about that class should be restricted to behavioral-probe status.

If a design rule repeatedly proves impossible to satisfy without unacceptable degradation of general architectural function, the rule should be revised or scoped.

HDD-ISA also adopts an anti-rescue principle. A new intervention or estimand should not be introduced merely because a previous experiment failed to produce the desired result.

Finally, when intervention leakage remains sufficiently large to prevent identification of the relevant causal contribution, the appropriate result is NI rather than an interpretive rescue.

19. Conclusion

HDD-ISA translates the functional hypotheses of HDD into architectural requirements, intervention interfaces, target-engagement procedures, controls, and reporting standards. Its contribution is methodological rather than ontological.

The central distinctions are:

prediction $\nRightarrow$ *causation*
causation $\nRightarrow$ *recurrence*
recurrence $\nRightarrow$ *self-reference*
self-reference $\nRightarrow$ *self-modeling*

and:

architectural access $\nRightarrow$ *target engagement* $\nRightarrow$ *causal identification*

Likewise:

module separability $\nRightarrow$ *causal separability*

An architecture may expose a mechanism without allowing its functional role to be identified. A behavioral effect may be real without uniquely identifying the mechanism that produced it. And two modules may be structurally distinct while remaining causally inseparable under the available interventions.

The relevant architectural virtue is therefore not the number of components labeled "memory," "self," or "self-model." It is whether competing explanations can be independently manipulated, their targets verified, their cross-effects measured, and their predicted intervention signatures compared.

The HDD Gap is consequently not a gap between architecture and intelligence. It is the gap between:

having a mechanism

and:

knowing, with causal evidence, what functional role that mechanism plays.

HDD-ISA specifies the architectural interfaces required to make that gap experimentally tractable.

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